

Unit 4

STATES OF MATTER

35 items · 84 marks

Syllabus sections in this unit

4.1 The gaseous state: ideal and real gases and $pV = nRT$

4.2 Bonding and structure

Unit 4: States of matter**1. M/J 26/P21/Q2(c)(ii)**

Context

2 The Contact process uses sulfur to produce sulfuric acid.**stage 1** Sulfur is burned in excess air.**stage 2** $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$ **stage 3** SO_3 reacts to produce H_2SO_4 .

Context: part (c) (introduction)

(c) CaCO_3 and O_2 are added to the waste gases from the Contact process before they escape. This prevents the release of SO_2 into the atmosphere.**(ii)** Use the ideal gas equation to calculate the volume of $\text{O}_2(\text{g})$ that reacts with 0.500 mol of SO_2 at 400°C and 101 kPa in equation 1.

Include units in your answer.

volume of $\text{O}_2(\text{g}) =$

units

[4]

2. M/J 26/P23/Q2(b)

Context

2 Group 18 elements exist as colourless gases at room temperature and pressure.**Table 2.1**

atom	relative volume of atom
neon	172
argon	287
krypton	355
xenon	446

(b) Identify **one** Group 18 element from Table 2.1 that behaves most like an ideal gas under standard conditions. Explain your answer.

[3]

3. M/J 26/P23/Q2(c)

Context

- 2** Group 18 elements exist as colourless gases at room temperature and pressure.

Table 2.1

atom	relative volume of atom
neon	172
argon	287
krypton	355
xenon	446

- (c) A sample of 1.775 g of gas **X** occupies 617.0 cm³ at 300 K and 101 kPa.

Calculate the relative molecular mass, M_r , of **X** assuming that **X** behaves as an ideal gas.

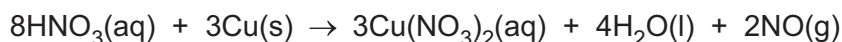
$$M_r \text{ of } \mathbf{X} = \quad [4]$$

4. MAR 26/P22/Q2(b)

Context

- 2** Nitrogen monoxide, NO, is an atmospheric pollutant.

- (b) In the laboratory, NO is made in the reaction of copper metal with dilute nitric acid. The reaction is kept cold to prevent the oxidation of NO.



A student reacts 0.00393 mol Cu with excess dilute HNO₃(aq) at 5 °C and 101 kPa.

Calculate the maximum volume, in cm³, of NO(g) that is produced under these conditions.

$$\text{volume of NO(g) produced} = \quad \text{cm}^3 [4]$$

5. O/N 25/P24/Q2(a)

Context

- 2** Aluminium oxide, Al₂O₃, and phosphorus(V) oxide, P₄O₁₀, are both used as reagents and catalysts.

- (a) The melting point of Al₂O₃ is 2072 °C. The melting point of P₄O₁₀ is 340 °C.

Explain the difference in the melting points of these two compounds.

[3]

6. M/J 25/P21/Q1(a)(i)

Context: part (a) (introduction)

- 1** (a) Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

- (i) Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium

bonding in sodium oxide

[1]

7. M/J 25/P21/Q1(a)(iii)

Context: part (a) (introduction)

- 1 (a) Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.
- (iii) Explain why sodium oxide conducts electricity when molten.

[1]

8. M/J 25/P21/Q1(d)

- (d) Table 1.2 shows melting points of some oxides.

Table 1.2

oxide	melting point/°C
SO ₂	-73
H ₂ O	0
SO ₃	17
SiO ₂	1610
MgO	2852
Al ₂ O ₃	2072

A student suggests the following hypothesis.

*The higher the oxidation number of the element combined with oxygen,
the higher the melting point of the oxide.*

Use Table 1.2 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

9. M/J 25/P22/Q1(a)

- 1 (a) Diamond and graphite are both crystalline solids made from carbon atoms. Graphite conducts electricity. Diamond does **not** conduct electricity.
- (i) Name the type of lattice structure present in the crystalline solids diamond and graphite.
- (ii) Explain how graphite conducts electricity.
- (iii) Explain why diamond does **not** conduct electricity.

[1]

[1]

[1]

10. M/J 25/P22/Q1(d)

(d) The melting points of different oxides are shown in Table 1.1.

Table 1.1

oxide	melting point / °C	force of attraction broken during melting
SO ₂	-73	
H ₂ O	0	
SiO ₂	1610	
Na ₂ O	1132	
MgO	2852	

- (i) Complete Table 1.1 by identifying the strongest force of attraction in each oxide that is broken during melting. Use the abbreviations below.

i.d. = instantaneous dipole–induced dipole

p.d. = permanent dipole–permanent dipole

H = hydrogen bond

C = covalent bond

I = ionic bond

[3]

- (ii) A student suggests the following hypothesis.

*The stronger the covalent bond between atoms in non-metal oxides,
the higher the melting point.*

Use Table 1.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

11. M/J 25/P23/Q2(c)(ii)

Context

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

Context: part (c) (introduction)

(c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point/°C
SO ₂	−73
H ₂ O	0
SO ₃	17
SiO ₂	1610
Na ₂ O	1132
MgO	2852
Al ₂ O ₃	2072

(ii) A student suggests the following hypothesis.

The melting point of an ionically bonded oxide is only determined by the charge on the cation.

Use Table 2.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

12. M/J 25/P24/Q1(a)

Context

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(a) Complete Table 1.1 by identifying the lattice structures of the crystalline solids of Mg, Si and P.

Table 1.1

element	Mg	Si	P
lattice structure in crystalline solid			

[3]

13. M/J 25/P24/Q6(b)

Context

6 P, Q and R are three different hydrocarbon molecules.

(b) Q is a volatile hydrocarbon.

0.194 g of gaseous Q occupies a volume of 71.5 cm³ at 100 °C and 100 kPa.

Use the ideal gas equation to calculate the M_r of Q.

 M_r of Q =

[2]

14. O/N 24/P21/Q3(a)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (a) (introduction)

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(ii) Identify the lattice structure shown by graphite.

[1]

15. O/N 24/P21/Q3(e)(iii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (e) (introduction)

(e) Silicon readily reacts with elements of high electronegativity.

(iii) SiO_2 is a white solid that melts above 1700°C .

SiCl_4 is a colourless liquid at room temperature.

Explain the difference in the melting points of these two compounds with reference to their structure and bonding.

[2]

16. O/N 24/P22/Q3(d)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.**(d)** Complete Table 3.1 to show the properties of nitrogen and phosphorus in their standard states.**Table 3.1**

	nitrogen	phosphorus
state and appearance of standard state	colourless gas	white solid
electrical conductivity		poor
type of bonding		
type of structure	simple	

[2]

17. O/N 24/P22/Q3(e)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.**(e)** A form of solid nitrogen has a lattice structure similar to solid iodine.

Identify the type of lattice structure of solid nitrogen.

[1]

18. O/N 24/P23/Q3(a)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (a) (introduction)

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.**Table 3.1**

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1.

[3]

(ii) Identify the lattice structure shown by graphite.

[1]

19. O/N 24/P23/Q3(e)(iii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (e) (introduction)

(e) Silicon readily reacts with elements of high electronegativity.

(iii) SiO_2 is a white solid that melts above 1700°C .

SiCl_4 is a colourless liquid at room temperature.

Explain the difference in the melting points of these two compounds with reference to their structure and bonding.

[2]

20. M/J 24/P21/Q2(d)(ii)

Context: part (d) (introduction)

(d) Boron nitride is a white solid that melts above 2900°C .

Fig. 2.2 shows part of the lattice structure of a crystal of boron nitride.

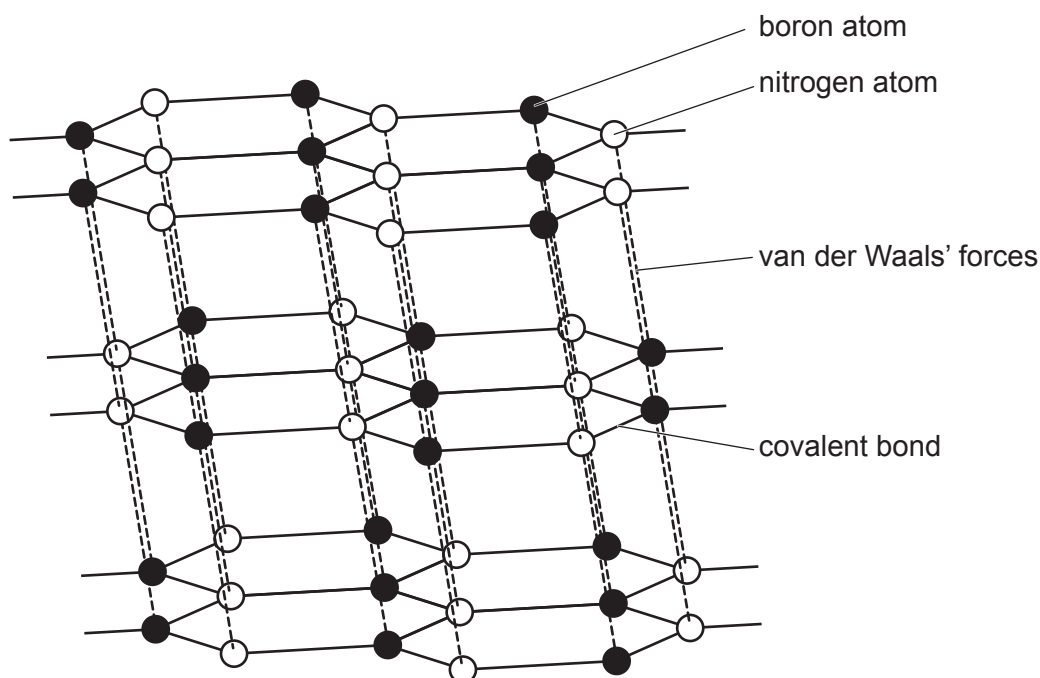


Fig. 2.2

(ii) Suggest the identity of another crystalline solid that has atoms arranged in layers similar to that of solid boron nitride.

[1]

21. M/J 24/P23/Q3(c)(i)

Context: part (c) (introduction)

(c) Boron carbide is a hard crystalline solid that has a melting point greater than 2000°C .

(i) Suggest the structure and bonding in boron carbide.

[1]

22. MAR 24/P22/Q1(b)(ii)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (b) (introduction)

- (b)** Bismuth reduces water to form bismuth oxide, Bi_2O_3 . A colourless gas that ignites with a squeaky pop also forms.
- (ii)** Bi_2O_3 is a yellow insoluble solid that melts at 1090 K. The molten compound conducts electricity.

Deduce the structure and bonding of Bi_2O_3 . Explain your answer.

[2]

23. O/N 23/P21/Q1(d)(i)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^-

Context: part (d) (introduction)

- (d)** POCl_3 shows similar chemical properties to PCl_5 .

POCl_3 has a melting point of 1°C and a boiling point of 106°C .

POCl_3 reacts vigorously with water, forming misty fumes and an acidic solution.

- (i)** Explain how the information in **(d)** suggests the structure and bonding of POCl_3 is simple covalent.

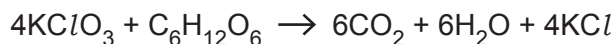
[2]

24. O/N 23/P21/Q3(d)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(d) Molten KClO_3 reacts with glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.



KClO_3 melts at 630 K. At this temperature, both CO_2 and H_2O are gases.

- (i)** Use the ideal gas equation to calculate the volume, in m^3 , of one mole of gas at 630 K and $1.00 \times 10^5 \text{ Pa}$.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = m^3
[1]

- (ii)** 5.00 g of $\text{C}_6\text{H}_{12}\text{O}_6$ reacts completely with molten KClO_3 .

Use your answer to **(d)(i)** to calculate the total volume of gas released at 630 K and $1.00 \times 10^5 \text{ Pa}$ in this reaction.

(If you were unable to answer **(d)(i)**, use 0.0463 m^3 in this question. This is **not** the correct answer to **(d)(i)**.)

total volume of gas released = m^3
[2]

25. O/N 23/P23/Q1(d)(i)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

Context: part (d) (introduction)

- (d)** POC_l₃ shows similar chemical properties to PC_l₅.

POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (i)** Explain how the information in **(d)** suggests the structure and bonding of POC_l₃ is simple covalent.

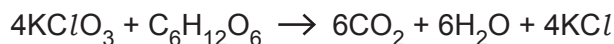
[2]

26. O/N 23/P23/Q3(d)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(d) Molten KClO_3 reacts with glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.



KClO_3 melts at 630 K. At this temperature, both CO_2 and H_2O are gases.

- (i) Use the ideal gas equation to calculate the volume, in m^3 , of one mole of gas at 630 K and $1.00 \times 10^5 \text{ Pa}$.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = m^3
[1]

- (ii) 5.00 g of $\text{C}_6\text{H}_{12}\text{O}_6$ reacts completely with molten KClO_3 .

Use your answer to (d)(i) to calculate the total volume of gas released at 630 K and $1.00 \times 10^5 \text{ Pa}$ in this reaction.

(If you were unable to answer (d)(i), use 0.0463 m^3 in this question. This is **not** the correct answer to (d)(i).)

total volume of gas released = m^3
[2]

27. M/J 23/P22/Q1(a)(i)-(ii)

Context

- 1** The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point / K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (a) (i) State the type of bonding present in magnesium and in sodium chloride.

bonding in magnesium

bonding in sodium chloride

[1]

- (ii) Explain the difference in the melting points of magnesium and sodium chloride.

[1]

28. M/J 23/P23/Q1(a)(i)

Context

- 1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

(a) (i) Identify the lattice structure of copper.

[1]

29. MAR 23/P22/Q1(e)(ii)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

Context: Table 1.2

Table 1.2

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment				

- (ii) At 150 °C and 103 kPa, all of the compounds listed in Table 1.2 are gases.

Under these conditions, 0.284 g of one of the compounds occupies a volume of 127 cm³.

Use this information to calculate the M_r of the compound. Hence, identify the compound from those given in Table 1.2.

Show your working.

 $M_r =$

identity of compound =

[3]

30. O/N 22/P22/Q2(d)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

- (d)** A 3.30 g sample of a Period 3 chloride is heated to 500 K in a sealed flask. At this temperature, the chloride is a gas of volume 250 cm^3 and the pressure in the flask is 323 kPa.

Use the ideal gas equation $pV = nRT$ to calculate the M_r of the Period 3 chloride. Deduce its formula.

$$M_r =$$

formula of Period 3 chloride =

[3]

31. O/N 22/P22/Q2(f)(ii)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

Context: part (f) (introduction)

- (f)** Bismuth is a dense metal in the same group as phosphorus.

- (ii)** Bismuth reacts with chlorine to form BiCl_3 .
 BiCl_3 is a solid at room temperature. It melts when heated gently.
 BiCl_3 reacts vigorously with water at room temperature to form an acidic solution.

Suggest the type of bonding and structure shown by BiCl_3 . Explain your answer.

[2]

32. M/J 22/P22/Q1(a)

- 1 (a) Magnesium has a melting point of 650°C and high electrical conductivity.

Explain these properties of magnesium by referring to its structure and bonding.

[2]

33. M/J 22/P22/Q2(a)

Context

- 2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl_2 , has a melting point of 900°C and is soluble in water.

- (a) Predict the lattice structure of $\text{RaCl}_2(\text{s})$ based on the properties described.

[1]

34. M/J 22/P23/Q1(f)(ii)

- (ii) Copper has a melting point of 1085°C and a high electrical conductivity.

Explain these properties of copper by referring to its structure and bonding.

[2]

35. MAR 22/P22/Q2(c)(i)

Context

- 2 Some oxides of elements in Period 3 are shown.



Context: part (c) (introduction)

- (c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

- (i) Deduce the type of structure and bonding shown by P_4O_6 . Explain your answer.

[2]

Answers Section

1. M/J 26/P21/Q2(c)(ii)

2(c)(ii)	M1 $PV=nRT$ M2 number of mol of O_2 required $n = 0.5 \div 2 = 0.25$ M3 temperature conversion to kelvin $= 400 + 273 = 673$ M4 $vol = (M2 \times 8.31 \times M3) \div 101\,000$ $= 0.0138431$ AND m^3	4 check the answer line first. M1 BOD upper / lower case discrepancies 0.0138(431) AND m^3 ✓✓✓✓ 13.8 and dm^3 OR 13 800 and cm^3 ✓✓✓✓ 0.014 and m^3 OR 14 and dm^3 or 14 000 and cm^3 ✓✓✓✓ 0.0276(86287) AND m^3 ✓✓✓ 0.00822(7722772) AND m^3 ✓✓✓✓ 0.0164(55445) AND m^3 ✓✓ A 2 sig figs up to calculator value final answers must be correctly rounded. M1–M3 may be seen in working if not described explicitly M1 A $V = nRT / P$ or A from working in M4 M3 A 273.15 OR 273.2 for 273 M3 A $400 + 273 / 673.15 / 673.2$ M4 volume calculated correctly using nRT / P AND correct / appropriate unit M4 A ecf from M2 and / or M3
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2. M/J 26/P23/Q2(b)

2(b)	M1 neon / Ne (most like an ideal gas) M2 it has the smallest volume M3 weakest vdw / temporary dipole – induced dipole attractions / London (dispersion) forces / London forces	3 M2 and M3 comparative term required once only e.g. Ne has smaller volume and weak vdw M2 BOD 'it is the smallest (atom)' M2 I mass BOD molecules for atoms intermolecular forces / IMF's = vdw M3 BOD least for weakest M3 A 'forces of attraction between atoms' I smallest number of electrons
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3. M/J 26/P23/Q2(c)

2(c)	M1 $PV=nRT$ M2 use of appropriate pressure value with volume e.g. convert volume to m^3 and kPa to Pa OR $101 \times 617.0 \times 10^{-3}$ etc. M3 calculate value for n $n = (101\,000 \times 617.0 \times 10^{-6}) \div 8.31 \times 300$ $= 0.024997 / 0.025$ M4 $(1.775 / M_r(X) = 0.02499 / 0.025)$ $M_r = 71(.0)$	4 Final answer 71(.0) ✓✓✓✓ M1 I subsequent incorrect rearrangement M2 A from calculations which produce consistent conversion for volume and pressure ie 101 and $617(.0) \times 10^{-3}$ (I ref to units for volume & pressure even if incorrect) M3 & M4 A ecf from incorrect conversion of volume and / or pressure units M3 A ecf from transcription errors / incorrect T using correct manipulation of $PV=nRT$ M4 A ecf from number of moles calculated M4 A ecf from M3 I units
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4. MAR 26/P22/Q2(b)

2(b)	M1 moles of NO = $0.00393 \times 2 / 3 = 0.00262$ M2 where $p = 101\,000$ (Pa) $R = 8.31$ $T = 278$ $n = 0.00262$ OR 101 (kPa) M3 use of $pV = nRT$ OR $V = nRT / p$ M4 volume = $5.99 \times 10^{-5} \times 10^6 = 59.9 \text{ cm}^3$ OR $5.99 \times 10^{-2} \times 10^3$	4
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5. O/N 25/P24/Q2(a)

2(a)	M1 Al_2O_3 is ionic M2 P_4O_{10} is a (simple) molecule M3 ionic bonds are stronger than intermolecular forces	3
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6. M/J 25/P21/Q1(a)(i)

1(a)(i)	sodium – metallic AND sodium oxide – ionic	1
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7. M/J 25/P21/Q1(a)(iii)

1(a)(iii)	(sodium and oxide) ions move (through the liquid)	1
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8. M/J 25/P21/Q1(d)

1(d)	<i>All three for two marks:</i> • false • comparison or statement of oxidation number • correct link to melting point from their comparison or statement	2
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9. M/J 25/P22/Q1(a)

1(a)(i)	(both) giant molecular	1
1(a)(ii)	delocalised electrons move through the structure OR delocalised electrons move through / between the layers	1
1(a)(iii)	no delocalised electrons (to carry charge through the structure) OR <u>all</u> (4) outer electrons are used in bonding	1

10. M/J 25/P22/Q1(d)

1(d)(i)	<table border="1"> <tbody> <tr> <td>SO_2</td><td>–73</td><td>pd</td></tr> <tr> <td>H_2O</td><td>0</td><td>H</td></tr> <tr> <td>SiO_2</td><td>1600</td><td>C</td></tr> <tr> <td>Na_2O</td><td>1132</td><td>I</td></tr> <tr> <td>MgO</td><td>2852</td><td>I</td></tr> </tbody> </table>	SO_2	–73	pd	H_2O	0	H	SiO_2	1600	C	Na_2O	1132	I	MgO	2852	I	3
SO_2	–73	pd															
H_2O	0	H															
SiO_2	1600	C															
Na_2O	1132	I															
MgO	2852	I															
1(d)(ii)	<i>Evaluate hypotheses based on the information candidates have given in the table</i> M1 not enough information M2 SO_2 and / or H_2O break intermolecular forces OR SO_2 and / or H_2O does not break covalent bonds OR only SiO_2 breaks covalent bonds	2															

11. M/J 25/P23/Q2(c)(ii)

2(c)(ii)	• false • Al_2O_3 melting point is lower than MgO • (but) charge on Al ion is greater than that on Mg Any 2 points = 1 mark , all 3 points = 2 marks	2
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12. M/J 25/P24/Q1(a)

1(a)	<table> <tr> <td><i>Mg</i></td><td><i>Si</i></td><td><i>P</i></td></tr> <tr> <td>M1 Giant metallic</td><td>M2 Giant molecular</td><td>M3 Simple molecular</td></tr> </table>	<i>Mg</i>	<i>Si</i>	<i>P</i>	M1 Giant metallic	M2 Giant molecular	M3 Simple molecular	3
<i>Mg</i>	<i>Si</i>	<i>P</i>						
M1 Giant metallic	M2 Giant molecular	M3 Simple molecular						

13. M/J 25/P24/Q6(b)

6(b)	M1 convert temp to 373 K (and use) M2 correct rearrangement and use $7.15 = n \cdot 8.31(4) \times T$ to find n AND $Mr = 0.194 / n$ $Mr = 84.1$	2
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14. O/N 24/P21/Q3(a)(i)-(ii)

3(a)(i)	<table> <tr> <td>element</td><td>C (graphite)</td><td>Sn</td></tr> <tr> <td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr> <tr> <td>electrical conductivity</td><td>• good/ conductor</td><td>good</td></tr> <tr> <td>type of bonding</td><td>• covalent</td><td>metallic</td></tr> <tr> <td>type of structure</td><td>giant</td><td>• giant</td></tr> </table>	element	C (graphite)	Sn	state and appearance	grey shiny solid	silvery solid	electrical conductivity	• good/ conductor	good	type of bonding	• covalent	metallic	type of structure	giant	• giant	3
element	C (graphite)	Sn															
state and appearance	grey shiny solid	silvery solid															
electrical conductivity	• good/ conductor	good															
type of bonding	• covalent	metallic															
type of structure	giant	• giant															
3(a)(ii)	giant molecular	1															

15. O/N 24/P21/Q3(e)(iii)

3(e)(iii)	M1 SiO_2 is giant covalent AND $SiCl_4$ is simple covalent / simple molecular / molecular M2 so need less energy to overcome IMF's in $SiCl_4$, compared to the bonds in SiO_2 ora	2
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16. O/N 24/P22/Q3(d)

3(d)	<table> <tr> <td>element</td><td>nitrogen</td><td>phosphorus</td></tr> <tr> <td>state and appearance</td><td>colourless gas</td><td>white solid</td></tr> <tr> <td>electrical conductivity</td><td>• poor</td><td>poor</td></tr> <tr> <td>type of bonding</td><td>• covalent</td><td>• covalent</td></tr> <tr> <td>type of structure</td><td>simple</td><td>• simple</td></tr> </table>	element	nitrogen	phosphorus	state and appearance	colourless gas	white solid	electrical conductivity	• poor	poor	type of bonding	• covalent	• covalent	type of structure	simple	• simple	2
element	nitrogen	phosphorus															
state and appearance	colourless gas	white solid															
electrical conductivity	• poor	poor															
type of bonding	• covalent	• covalent															
type of structure	simple	• simple															

17. O/N 24/P22/Q3(e)

3(e)	simple molecular (lattice structure)	1
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18. O/N 24/P23/Q3(a)(i)-(ii)

3(a)(i)	element	C (graphite)	Sn	3
	state and appearance	grey shiny solid	silvery solid	
	electrical conductivity	• good/ conductor	good	
	type of bonding	• covalent	metallic	
	type of structure	giant	• giant	
3(a)(ii)	giant molecular			1

19. O/N 24/P23/Q3(e)(iii)

3(e)(iii)	M1 SiO₂ is giant covalent AND SiCl₄ is simple covalent / simple molecular / molecular M2 so need less energy to overcome IMF's in SiCl₄, compared to the bonds in SiO₂ ora	2
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20. M/J 24/P21/Q2(d)(ii)

2(d)(ii)	graphite	1
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21. M/J 24/P23/Q3(c)(i)

3(c)(i)	giant (molecular and) covalent	1
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22. MAR 24/P22/Q1(b)(ii)

1(b)(ii)	giant AND has a high melting point	1
	ionic AND conducts electricity (only) when molten	1

23. O/N 23/P21/Q1(d)(i)

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	1
	(vigorous reaction with water suggests) hydrolysis	1

24. O/N 23/P21/Q3(d)

3(d)(i)	$pV = nRT \therefore V = 8.31 \times 630 \div 100\,000 = \underline{\underline{0.0524}}$ (m ³)	1
3(d)(ii)	moles of gas = $12 \times 5.00 / 180$	1
	volume of gas = $M1 \times \text{(d)(i)} = \underline{\underline{0.0175}}$ (m ³)	1

25. O/N 23/P23/Q1(d)(i)

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	1
	(vigorous reaction with water suggests) hydrolysis	1

26. O/N 23/P23/Q3(d)

3(d)(i)	$pV = nRT \therefore V = 8.31 \times 630 \div 100\,000 = \underline{\underline{0.0524}}$ (m ³)	1
3(d)(ii)	moles of gas = $12 \times 5.00 / 180$	1
	volume of gas = $M1 \times \text{(d)(i)} = \underline{\underline{0.0175}}$ (m ³)	1

27. M/J 23/P22/Q1(a)(i)-(ii)

1(a)(i)	bonding in magnesium – metallic AND bonding in sodium chloride – ionic	1
1(a)(ii)	bonds in NaCl are stronger than bonds in Mg	1

28. M/J 23/P23/Q1(a)(i)

1(a)(i)	giant metallic	1
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29. MAR 23/P22/Q1(e)(ii)

1(e)(ii)	conversion of units 103000 Pa $127 \times 10^{-6} \text{ m}^3$ 423 K	1
	Use of $pV = (m / M_r)RT$ $M_r = \frac{0.284 \times 8.31 \times 423}{103000 \times 127 \times 10^{-6}}$	1
	$M_r = 76.3$ AND compound = CS ₂	1

30. O/N 22/P22/Q2(d)

2(d)	M1 value of n based on re-arranged equation for 'n' and substitution of correct values $n = \frac{3.23 \times 10^5 \cdot 250 \times 10^{-6}}{8.31 \cdot 500} = 0.0194(34416)$	1
	M2 calculated value of M_r based on M1 $M_r = 3.30 \div n = 170$	1
	M3 period 3 chloride which matches M_r in M2 (formula) SiCl ₄	1

31. O/N 22/P22/Q2(f)(ii)

2(f)(ii)	M1 (structure =) simple/molecular, because it has a low melting/boiling point <i>owtte</i>	1
	M2 (bonding =) covalent, because it is hydrolysed	1

32. M/J 22/P22/Q1(a)

1(a)	M1 comment explaining high melting point of Mg in terms of many strong metallic bonds OR many strong (electrostatic) attractions between cations and delocalised electrons OR strong bonds in giant metallic structure. M2 comment explaining electrical conductivity of Mg in terms of movement of delocalised electrons delocalised electrons can move through the structure	2
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33. M/J 22/P22/Q2(a)

2(a)	giant ionic	1
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34. M/J 22/P23/Q1(f)(ii)

1(f)(ii)	M1 comment explaining high melting point of Cu many strong metallic bonds OR many strong (electrostatic) attractions between cations and delocalised electrons OR strong bonds in giant metallic structure. M2 comment explaining electrical conductivity of Cu delocalised electrons are free to move through the structure (<i>owtte</i>)	2
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35. MAR 22/P22/Q2(c)(i)

2(c)(i)	(structure =) simple/molecular, because it has a low melting/boiling point [1] (bonding =) covalent, because it is hydrolysed [1]	2
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